Kyle A. Moore

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SUMMARY

AI/ML researcher completing a Ph.D. at Vanderbilt University with published work at top AI conferences, including EMNLP, AAAI, CoNLL, and CogSci. My research specialty is the behavior, uncertainty, and cognitive alignment of large language models with an eye toward effective LLM evaluation, reasoning, and interpretability. Designed and independently taught university-level AI coursework; mentored undergraduate research teams from problem formulation through publication.

RESEARCH INTERESTS

Artificial Intelligence & Machine Learning · Natural Language Processing · Large Language Models · AI Behavior & Human–AI Comparison · Human-AI Interaction (future direction) · Affective Computing (future direction)

EDUCATION

Ph.D., Computer Science – Vanderbilt University *Aug 2019 – Dec 2026*
Ph.D. completed; degree to be awarded Dec 2026
Dissertation: "Human-Analogous Uncertainty Behavior: Cognitive Studies of Large Language Models"
Advisors: Doug Fisher, Meiyi Ma

M.S., Computer Science – University of Mississippi *Aug 2017 – May 2019*
Thesis: "Building an Automated QA System Using Online Forums as Knowledge Bases "
Advisor: Naeemul Hassan

B.S., Computer Science – University of Mississippi *Aug 2013 – May 2017*

PUBLICATIONS

Peer-Reviewed

- Fisher, D., **Moore, K.**, Roberts, J. "State-Space Search and Transformer-Based AI Hybrids." ACS 2026.
- **Moore, K.**, Roberts, J., Watson, D., Wisniewski, P. "Investigating Human-Aligned Large Language Model Uncertainty." FLAIRS 2026.
- Sawyer, H., Roberts, J., **Moore, K.** "Basic Categories in Vision Language Models: Expert Prompting Doesn't Grant Expertise." ACS 2025.
- **Moore, K.**, Roberts, J., Pham, T., Fisher, D. "Chain of Thought Still Thinks Fast: APriCoT Helps with Thinking Slow." CogSci 47, 2025.
- **Moore, K.**, Roberts, J., Pham, T., Ewaleifoh, O., Fisher, D. "The Base-Rate Effect on LLM Benchmark Performance: Disambiguating Test-Taking Strategies from Benchmark Performance." EMNLP 2024.
- Roberts, J., **Moore, K.**, Fisher, D., Ewaleifoh, O., Pham, T. "Large Language Model Recall Uncertainty is Modulated by the Fan Effect." CoNLL 28, 2024.
- Roberts, J., **Moore, K.**, Wilenzick, D., Fisher, D. "Using Artificial Populations to Study Psychological Phenomena in Neural Models." AAAI 38, 2024.
- Timalsina, U., Broll, B., **Moore, K.**, Lédeczi, Á. "DeepForge for Astronomy: Deep Learning SDSS Redshifts from Images." Astronomy and Computing 40 (2022).

Workshop Papers

- Roberts, J., **Moore, K.**, Fisher, D. "Do Large Language Models Learn Human-Like Strategic Preferences?" REALM Workshop, ACL 2025.

Preprints/Under Review

- **Moore, K.**, Roberts, J., Watson, D., Ward, W., Heyboer, G. "Human-Alignment, Calibration, and Activation Patterns in Large Language Model Uncertainty." Preprint, under review, 2026.

RESEARCH EXPERIENCE

Graduate Research Assistant – Vanderbilt University Aug 2019 – Present

- Designed and ran controlled behavioral studies testing whether LLMs exhibit human-analogous patterns of uncertainty behaviors like typicality and fan effects, bridging cognitive science methodology and NLP evaluation.
- Showed that answer-token base rates, not knowledge, drive much of reported MMLU accuracy across 5 open LLMs; introduced Nvr-X, a dataset-agnostic augmentation procedure that disentangles task performance from test-taking strategy in MCQA benchmarks, under which only 1 of 5 models beat random chance.
- Established the first direct human-vs-LLM uncertainty comparison on paired tasks, scaling to 30 open-weight models and 4 QA datasets; found that instruct fine-tuning degrades both human-alignment and calibration, and that internal activation probes recover substantially more alignment signal than output logits.
- Mentored 6 undergraduate researchers from problem formulation through publication, co-authoring 6 papers.

TEACHING EXPERIENCE

Instructor - CS 4260: Introduction to Artificial Intelligence – Vanderbilt University Spring 2025

- Designed the course and taught it as sole lecturer to ~60 undergraduate and 2 graduate students, with full responsibility for syllabus, materials, assessment, and grading. Graduate students are not ordinarily appointed to teach at Vanderbilt; the arrangement was documented by a letter from the Dean. Course listed under faculty advisor Doug Fisher, who held the formal appointment. More details, including said letter, are available on request.
- Covered topics include: goal-directed search, adversarial search, automated theorem proving, logic-based planning, CSPs, decision trees, neural networks, clustering, Bayesian probability, belief nets, (hierarchical hidden) Markov models, MDPs, RL

Teaching Assistanceships – Ph.D. – Vanderbilt University (* - guest lectures given)

- CS 4260: Artificial Intelligence (2022F, 2023F, 2024S*, 2025F*, 2026S*)
- CS 1151: Computer Ethics (Summer 2023, Summer 2024, Summer 2025)
- CS 3252: Theory of Automata, Formal Languages, and Computation (2021F, 2023S*, 2024F*)
- CS 3265: Databases (Summer 2022, Summer 2023)
- CS 3891: Computational Creativity (2023S*)
- CS 3892: Software for Autonomous Vehicles (2022S)
- CS 3270: Programming Languages (2019F, 2020S, 2020F. 2021S)

RESEARCH PRESENTATIONS

- Poster, AAAI 2024
- Poster, EMNLP 2024
- Poster, CogSci 2025
- Talk, FLAIRS 2026
- Talk, ACS 2026

PROFESSIONAL SERVICE

Peer Review

- ICLR 2026
- EMNLP 2026
- FLAIRS 2026
- AAAI 2026
- ACS 2025
- ACL Rolling Review (ARR), February 2025
- REALM Workshop, ACL 2025

Organizing Committee

- FLAIRS 2026 Special Track: Human-AI Collaboration and Augmented Intelligence (HAICAI)

Other Scholarly Contributions

- Acknowledged for pre-publication feedback on Anderson & Fisher, Artificial Intelligence for Academic Libraries (2025 Routledge)

AWARDS & HONORS

- IBM Goldstine Graduate Fellowship, Vanderbilt University

MENTORING & ADVISING

Undergraduate Research Intern Mentees (includes only mentees who earned authorship on a resulting paper, published or under review):

- Drew Wilenzick (AAAI 2024)
- Oseremhen Ewaleifoh (CoNLL 2024; EMNLP 2024)
- Thao Pham (CoNLL 2024; EMNLP 2024; CogSci 2025)
- Daryl Watson (FLAIRS 2026; preprint 2026)
- William Ward (preprint 2026)
- Greyson Heyboer (preprint 2026)

TECHNICAL SKILLS

Languages: Python (primary), Java, C/C++, SQL, Haskell, LaTeX
ML/Data: NumPy, pandas, Matplotlib, TensorFlow, PyTorch, scikit-learn, HF transformers
Methods: uncertainty quantification and calibration, activation probing, nonparametric statistics
(Natural) Languages: Japanese (Elementary Proficiency)

GRADUATE COURSES TAKEN

Ph.D. – Vanderbilt University

- CS 8395 – Special Topics: Computational Game Theory (2021F)
- CS 6388 – Model-Integrated Computing (2020F)
- CS 5891 – Special Topics: Social Network Analysis (2020F)
- CS 8395 - Special Topics: Visual Analytics and Machine Learning (2020S)
- CS 6360 – Advanced Artificial Intelligence (2020S)
- EECE 6354 – Advanced Real-Time Systems (2019F)
- CS 8395 – Special Topics: Computation and Cognition (2019F)
- CS 6376 – Hybrid and Embedded Systems (2019F)

M.S. – University of Mississippi

- CSCI 556 – Multiparadigm Programming (2018F)
- CSCI 561 – Computer Networks (2018F)
- CSCI 547 – Digital Image Processing (2018F)
- CSCI 581 – Special Topics: Compiler Design (2018S)
- CSCI 581 - Special Topics: Natural Language Processing (2018S)
- CSCI 531 - Artificial Intelligence (2018S)
- ENGR 692 – Special Topics: High-Performance Computing (2017F)
- ENGR 691 – Special Topics: Random Algorithms (2017F)
- CSCI 533 – Analysis of Algorithms (2017F)